

Saminur Islam

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SUMMARY

Computer Science PhD candidate researching agentic AI systems, LLMs, and behavioral modeling. My work spans building AI agents that make sequential decisions over real user data, fine-tuning and evaluating large language models, and applying deep learning to large-scale structured and unstructured data. I have industry experience deploying ML systems at AT&T Labs across 10,000+ sites and a paper on correcting training-time failure modes in self-training models under review at NeurIPS 2026. Strong in PyTorch, Hugging Face Transformers, LLM fine-tuning, representation learning, and large-scale ML experimentation.

EDUCATION

- **North Carolina State University** Raleigh, NC, USA
Ph.D. in Computer Science (Advisor: Dr. Collin Lynch) Aug 2022 - Aug 2027 (Expected)
- **West Virginia University** Morgantown, WV, USA
M.Sc. in Computer Science (Advisor: Dr. Donald Adjero) Aug 2020 - May 2022
- **Bangladesh University of Engineering and Technology** Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering Feb 2011 - Mar 2016

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, C, SQL, Bash Scripting, Go, Scala
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, representation learning, VAE, GAN, anomaly detection, time-series modeling, recommender systems, user behavior modeling
- **LLMs:** Transformer architectures, LLaMA, BERT, GPT-4, Gemini, fine-tuning, prompt engineering, RAG, LangChain
- **Data & Infrastructure:** PostgreSQL, MySQL, Redis, DynamoDB, AWS, Docker, Kubernetes, Git, Linux, real-time data pipelines, cloud storage, experiment tracking

RESEARCH AND INDUSTRY EXPERIENCE

- **North Carolina State University** Raleigh, NC
Graduate Assistant (Advisor: Dr. Collin Lynch) Aug 2022 - Present
 - Designed an agentic AI framework that models user behavioral trajectories over long sessions and learns when and how to intervene, framing adaptive decision-making as a sequential optimization problem over longitudinal interaction data.
 - Built and evaluated LLM-based feedback systems using fine-tuning, prompt engineering, and RAG, studying how language model outputs can be grounded in real behavioral context rather than static inputs.
 - Developed unsupervised clustering and representation learning pipelines over large-scale IDE interaction data to surface latent behavioral patterns across user cohorts.
 - Built evaluation pipelines to measure and audit agent decision quality across cohorts, tracking alignment between model outputs and ground-truth behavioral signals.
- **AT&T Labs** Bedminster, NJ
Research Intern Jun 2025 - Aug 2025
 - Designed and evaluated deep learning models for multivariate time-series monitoring across 10,000+ telecom nodes within AT&T's E911 critical network infrastructure.
 - Deployed a real-time anomaly detection pipeline that replaced a 3-day manual inspection process, reducing time-to-resolution to under an hour while achieving a 95% recall rate and a 63% reduction in false alarms.
 - Developed unsupervised representation learning techniques to decouple short-term anomalies from long-term performance drift, isolating latent degradation states before customer impact.
 - Built observability and alerting tools that let engineers trace the root cause of service degradation in near real time, cutting typical investigation time from days to under an hour and reducing false alarms by roughly 63 percent while catching around 95 percent of true failures.
- **West Virginia University** Morgantown, WV
Graduate Research Assistant (Advisor: Dr. Donald Adjero) Jan 2020 - May 2022
 - Designed a multimodal deep learning framework for drug-drug interaction discovery, publishing the peer-reviewed methodology and findings in Scientific Reports.
 - Fused heterogeneous protein sequence data and structure-derived features within similarity networks, achieving up to 98% precision and 99% AUC in predicting novel interactions.
 - Generated over 32,000 candidate interactions and engineered a thresholding pipeline to isolate 26,000 high-confidence predictions, with over 60% scoring above a 0.99 confidence metric.

- Graph-modeled latent biochemical relationships across multi-source biomedical datasets to successfully surface previously unmapped adverse drug reactions.

- **Highmark Health**

Pittsburgh, PA

Graduate Intern

May 2021 - Aug 2021

- Designed, coded, tested, and implemented a secure file transfer application enhancing data security and efficiency.
- Worked with interdisciplinary teams to analyze complex data needs and delivered solutions that streamlined process efficiency and communication.

- **LR Global Bangladesh Ltd**

Dhaka, Bangladesh

Software Developer

Mar 2018 - Dec 2019

- Built end-to-end data pipelines for real-time ingestion, processing, and storage of financial time-series data, supporting automated decision-making in investment workflows at scale.
- Maintained cloud infrastructure on AWS and used Redis for low-latency in-memory caching, keeping model inference and analytics pipelines responsive under production load.
- Developed statistical and ML models over financial time-series data and deployed them through scalable backend systems integrated with the existing data pipeline.

- **Synesis IT**

Dhaka, Bangladesh

Programmer

Nov 2016 - Dec 2018

- Built full-stack applications across Java, Python, .NET, and PHP, choosing the right stack for each project's performance and scaling needs.
- Delivered web systems for high-profile clients including Bangladesh Police, Bangladesh Post Office, and major telecom operators, streamlining their core operational workflows.
- Designed and optimized relational database schemas, tuning complex queries to improve throughput and maintain data integrity under production load.
- Worked directly with government and telecom stakeholders to translate messy, ambiguous requirements into clean technical specs, use cases, and process diagrams.

SELECTED PUBLICATIONS

[View profile on Google Scholar →](#)

- **Conference Proceedings**

- * **S. Islam**, Z. Gao, and C. Lynch, "PALMU: Correcting the Hidden Cost of Forgetting in Self-Training Models," (Submitted to NeurIPS 2026). Fixes hidden pseudo-label corruption caused by machine unlearning in self-training models without full retraining, addressing a failure mode in the same self-improvement loops that underlie Agentic AI.
- * S. Gilson, T. Barnes, M. Tamanna, and **S. Islam**, "How Do Undergraduates Interested in Computer Science Use Online, On-Demand CS Coursework?" *Proceedings of the Twelfth ACM Conference on Learning & Scale (L&S '25)*, pp. 271–275. doi:10.1145/3698205.3733938.
- * A. Limke, **S. Islam**, B. Riahi, X. Tian, M. Hill, V. Cateté, and T. Barnes, "What Does It Take to Support Problem Solving in Programming Classrooms? A New Framework from the K-12 Teacher Perspective," *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, Art. 591, pp. 1–7. doi:10.1145/3706599.3719763.

- **Journal Papers**

- * **S. Islam**, A. Abbasi, N. Agarwal, W. Zheng, G. Doretto, and D. A. Adjeroh, "PS3N: Leveraging Protein Sequence-Structure Similarity for Novel Drug-Drug Interaction Discovery," *Scientific Reports*, vol. 15, no. 37191, 2025. <https://doi.org/10.1038/s41598-025-14690-w>.
- * **S. Islam**, Z. Gao, S. Heckman, and C. Lynch, "Characterizing Student Coding Behaviors from IDE Traces: Unsupervised Clustering of Code Evolution Metrics," *Under Review: Journal of Educational Data Mining*.
- * **S. Islam**, Z. Gao, V. Patwardhan, S. Heckman, and C. Lynch, "Debugging Around Help: An Episode-Level Analysis of Debugging and Help-Seeking Behaviors in Introductory Programming," *Under Review: Under Review: Journal of Educational Data Mining*.

- **Posters**

- * **S. Islam**, Z. Gao, J. Bacher, G. Silva De Oliveira, V. Patwardhan, S. Heckman, and C. Lynch, "EclipseMonitor: A Real-Time Student Programming Environment Data Collection Tool," *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE '25)*, pp. 1489–1490. doi:10.1145/3641555.3705170.
- * Z. Gao, **S. Islam**, C. Scott, C. Lynch, and S. Heckman, "Comparing Students' and Teachers' Assessments of Office Hours," *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE '25)*, pp. 1459–1460. doi:10.1145/3641555.3705147.